Hyojoon Park

PH.D. CANDIDATE | RESEARCH SCIENTIST (3D GRAPHICS, VISION, DEEP LEARNING)

hyojoon.park@wisc.edu | Personal Website | Github | Linkedin | Google Scholar

Professional Summary

- My research integrates **Machine Learning** with **Computer Graphics** and **Computer Vision**, building end-to-end systems that combine physics-based simulation, geometry, and image/video processing.
- My work spans **3D reconstruction** of bodies, faces, and scenes, from camera calibration and bundle adjustment to **3DGS**, along with **neural simulation** of face and hair, hand reconstruction for **XR (VR/AR)**, and **haptics** for dental simulation.
- I develop **deep learning-based 4D medical image and video compression** using spatiotemporal wavelets, hyperprior-based entropy modeling, latent quantization, and rate-distortion optimization.

Education

University of Wisconsin-Madison *Wisconsin, USA*
Ph.D. Candidate in Computer Sciences *Sep. 2021 - Expected Dec. 2026*
- **Research area:** Synergistic integration of machine learning in physics-based simulation for computer graphics and medical imaging
- **Advisor:** Professor Eftychios Sifakis, **Co-advisor:** Professor Kevin Eliceiri

Seoul National University *Seoul, Korea*
M.S. in Mechanical Engineering *Mar. 2017 - Feb. 2019*
- **Research area:** Rendering and transparent control of high-performance haptic system
- **Advisor:** Professor Dongjun Lee
- **Thesis:** Dental Simulator with Increased Z-width of Haptic Rendering (also presented at AsiaHaptics 2018)

  Technical Contributions
  - Extended passive decomposition to SE(3) rigid-body motion with passive midpoint integration for stable, stiff haptic rendering.
  - Derived a momentum-based force observer to estimate user-applied forces on multi-DoF haptic devices without force sensors.
  - Implemented octree-based voxel rendering for real-time collision detection and material removal in virtual tooth drilling.
- Selected for **Outstanding MS Thesis Presentation Award** [M.S. thesis presentation]

Technical University of Munich (TUM) *Munich, Germany*
B.S. Exchange Student in Mechanical Engineering *Spring 2014*

Korea University *Seoul, Korea*
B.S. in Mechanical Engineering *Mar. 2010 - Feb. 2017*
- Military service: Jun. 2011 - Mar. 2013

Work Experience

Google *Playa Vista, CA, USA*
Student Researcher *Sep. 2025 - May 2026*
- Developed a real-time, self-supervised neural hair simulator for digital humans, focusing on efficient physics-based animation for XR. Publication-track results in progress.

  Technical Contributions
  - Designed a GRU-based motion encoder to retain motion history and capture hair inertia, including natural swinging and settling.
  - Encoded local and global hair groom structure with a latent transformer, scaling linearly rather than quadratically in strand count.
  - Built a StyleGAN2-inspired modulated MLP decoder conditioned on body shape, hairstyle, and motion history for hair animation.
  - Trained the neural hair simulator with volumetric elastic, inertial, and collision losses for richer dynamics and faster training.

Nokia Bell Labs *New Providence, NJ, USA*
Industrial Metaverse Intern *Jun. 2025 - Aug. 2025*
- Developed a real-time 3DGS digital twin with precise physical registration (scale and orientation) and language features for natural-language object queries, as part of "Empowering Digital Twins with Precise Physical Registration and Language-Awareness." Selected as the **1st place winner** among 100+ global interns.
- Follow-up work studies how capture geometry affects 3DGS reconstruction quality. Publication-track results in progress.

  Technical Contributions
  - Aligned the digital twin with real-world coordinates using ChArUco board correspondences, linear triangulation, nonlinear reprojection-error minimization, and Umeyama similarity alignment.
  - Embedded language features in each 3D Gaussian and applied DBSCAN clustering for real-time, text-guided 3D object segmentation.

NVIDIA *Santa Clara, CA, USA*

Software Intern - Physics-Based Simulations *May 2024 - Aug. 2024*

- Developed a single-image 3D face reconstruction framework trained on synthetic data to generalize from infancy to old age.
- Enabled identity-preserving re-aging and de-aging using explicit age control, differentiable rendering, and adversarial training.

Technical Contributions

- Generated synthetic 3D face meshes from infancy to old age by transferring identity-specific deformations from scanned faces to age-specific face templates.
- Learned the 3DMM identity basis using PCA on skull-aligned face meshes centered on age-specific reference shapes.
- Trained an end-to-end network to predict 3DMM identity and age parameters from a single image, using the pretrained MiVOLO age estimator to supply an age prior.

University of Utah *Salt Lake City, UT, USA*

Graduate Research Assistant *Sep. 2019 - Jun. 2021*

- Developed and open-sourced a multi-view camera calibration framework (bundle adjustment, ML-based outlier detection) for "Capturing Detailed Deformations of Moving Human Bodies" (SIGGRAPH 2021), tracking 1,000+ uniquely labeled points to reconstruct fine skin deformations of a moving body using a low-cost patterned suit and occlusion-resilient deep networks. [code] [paper]

Technical Contributions

- Built an end-to-end C++ and Python pipeline for multi-camera calibration using checkerboard corner detection and bundle adjustment with Ceres Solver.
- Built a VAE-based outlier detector to filter erroneous checkerboard corners and improve camera calibration.
- Trained a CNN corner detector for a body reconstruction pipeline that rejects invalid image patches, recognizes two-letter codes on the capture suit, and triangulates labeled corners.

Korea Institute of Science and Technology (KIST) *Seoul, Korea*

Research Intern *Mar. 2019 - Aug. 2019*

- Developed a VR hand-rendering framework for wearable fingertip-tracking devices, reconstructing realistic full-hand poses from fingertip positions alone through motion retargeting and novel IK hinge constraints. Published as "Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices" (IROS 2020). [paper] [video]

Technical Contributions

- Calibrated virtual hand size using fingertip measurements, tracking-device geometry, and SVD-based similarity alignment.
- Retargeted finger joint angles using per-joint scaling and offsets calibrated from open-hand and grasping poses to align rendered and intended hand poses.
- Extended the FABRIK IK solver with planar hinge constraints to prevent unnatural finger bending during real-time hand reconstruction.

Publications

Near-realtime Facial Animation by Deep 3D Simulation Super-Resolution
Hyojoon Park, *Sangeetha Grama Srinivasan, Matthew Cong, Doyub Kim, Byungsoo Kim, Jonathan Swartz, Ken Museth, Eftychios Sifakis*
ACM Transactions on Graphics (TOG), 2024 (Presented at SIGGRAPH ASIA 2024) [paper] [code] [featured video]
- Upsamples coarse physics-based facial simulations with a neural network for detailed animation at 18 FPS (115x speedup).

Collagen Fiber Centerline Tracking in Fibrotic Tissue via Deep Neural Networks with Variational Autoencoder-based Synthetic Training Data Generation
Hyojoon Park*, *Bin Li*, Yuming Liu, Michael S. Nelson, Helen M. Wilson, Eftychios Sifakis, and Kevin W. Eliceiri (*equal contributions)*
Medical Image Analysis, 2023 [paper] [code]
- Introduces DuoVAE, pairing a main and auxiliary VAE to generate controllable synthetic data for collagen-fiber centerline tracking.

Capturing Detailed Deformations of Moving Human Bodies
*He Chen, **Hyojoon Park**, Kutay Macit, and Ladislav Kavan*
SIGGRAPH, 2021 [paper] [code]
- Reconstructs 1,000+ labeled 3D points using CNN corner detection, capture-suit code recognition, and multi-view triangulation.

Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices
Hyojoon Park *and Jung-Min Park*
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020 [paper] [video]
- Uses motion retargeting and FABRIK IK with planar hinge constraints for real-time full-hand reconstruction.

Publications (continued)

Dental Simulator with Increased Z-width of Haptic Rendering
***Hyojoon Park**, Myungsin Kim, and Dongjun Lee*
AsiaHaptics, 2018 [paper] [video]
- Combines passive midpoint integration and passive decomposition to increase stable haptic rendering stiffness by up to 10x.

Rigid-body Collaborative Manipulation among Remote Users with Wearable Cutaneous Haptic Interfaces
*Myungsin Kim, WonHa Lee, **Hyojoon Park**, Junghan Kwon, Yong-Lae Park, and Dongjun Lee*
AsiaHaptics, 2018 [paper] [video]
- Enables remote users to jointly manipulate rigid bodies through wearable cutaneous haptic interfaces, using passivity-based control for stable and realistic interactions.

Stretchable Skin-Like Cooling/Heating Device for Reconstruction of Artificial Thermal Sensation in Virtual Reality
*Jinwoo Lee, Heayoun Sul, Wonha Lee, Kyung Rok Pyun, Inho Ha, Dongkwan Kim, **Hyojoon Park**, Hyeonjin Eom, Yeosang Yoon, Jinwook Jung, Dongjun Lee, and Seung Hwan Ko*
Advanced Functional Materials, 2020 [paper]
- Reproduces both hot and cold sensations in VR from a single stretchable, skin-like device (230% stretchable), combining heating and cooling in one structure.

Highly Stretchable and Oxidation-resistive Cu Nanowire Heater for Replication of the Feeling of Heat in a Virtual World
*Dongkwan Kim, Junhyuk Bang, Wonha Lee, Inho Ha, Jinwoo Lee, Hyeonjin Eom, Myungsin Kim, Jungjae Park, Joonhwa Choi, Jinhyung Kwon, Seungyong Han, **Hyojoon Park**, Dongjun Lee, and Seung Hwan Ko*
Journal of Materials Chemistry A, 2020 [paper]
- Replicates virtual heat on a wearable glove using a 12-pixel stretchable thermal array with laser-patterned copper-nanowire heaters, controlling temperature to within 5% error.

Wearable Cutaneous Haptic Interface with Soft Sensors and IMUs
*Yongjun Lee, Myungsin Kim, Yongseok Lee, **Hyojoon Park**, and Dongjun Lee*
Korea Robotics Society Annual Conference, 2018

Design and Performance Evaluation of Wearable Haptic Interfaces
*WonHa Lee, Myungsin Kim, **Hyojoon Park**, and Dongjun Lee*
International Conference on Control, Automation and Systems, 2018

Services

SIGGRAPH Asia 2026 Reviewer *2026*

Technical Skills

Languages	Python, C++, C, CUDA
Frameworks	PyTorch, JAX, OpenCV, NumPy/SciPy, OpenGL
Core Areas	3D Vision, Neural Rendering, Physics-based Simulation, Deep Learning, Computer Graphics, 3D Reconstruction, Data Compression
Tooling	Linux, Git, Docker, CMake, LaTeX

Teaching Assistant

Computer Graphics (CS559) *University of Wisconsin-Madison* *Spring 2022*

Computer Graphics (CS559) *University of Wisconsin-Madison* *Fall 2021*

Interactive Computer Graphics (CS6610) *University of Utah* *Spring 2021*

Computer Graphics (CS4600) *University of Utah* *Fall 2020*

System Analysis in Mechanical Engineering *Seoul National University* *Spring 2018*